Higher Salaries Mean Greater Benefits: From SF's City Employees in 2024*

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Abstract

This study looks into how an employee's total benefits received is related to the their total salary. The objective was to identify any correlation between employees' salaries and the benefits that they received. For many employees on the market for a job, this is an important question because benefits make up a significant portion of overall compensation and relating it to salary would reveal how much an employer values them. Data regarding the total compensation of different employees was studied using employee data from the City of San Francisco for the year of 2024. Methods utilized included building a linear regression model, running a hypothesis test, as well as utilizing a scatterplot. Results from our hypothesis test concluded that the amount of benefits an employee receives is positively correlated to the total salary they receive.

Introduction

The amount of compensation employees obtain is crucial when it comes to workplace dynamics as well as employee retention. How well an employee is compensated is also directly related to how satisfied they are with their job. However, compensation is not just about the base salary an employee obtains. Benefits, such as health insurance, retirement, etc., are also a major factor. When deciding on a job offer, many prospective employees focus primarily on the salary as the key factor in evaluating the offer. Most job posts would also list a range for the salary, while completely leaving out details on benefits given. From a perspective employee's point of view, understanding the relation between their salary and their benefits would give another insight when evaluating offers. One would expect that the higher the salary an employee receives, the greater benefits they receive, simply because the higher salary would indicate that their labor is valued more by the employer.

*Project GitHub Repository: https://github.com/TFang96/MATH261A

Understanding the relationship between total salary and total benefits would also reveal how corporations allocate resources across employees of different roles. If we conclude that benefits and salary are not correlated with each other, it would show that the benefits corporations give to their employees are not determined by their role in the company, since the role determines salary. In contrast, if the amount of benefits an employee obtains is dependent on the total salary they receive and there is a positive correlation, this would point to the fact that organizations not only give more compensation to employees of certain levels or roles through their salary, but the recognition is reflected in their benefits as well.

This paper addresses these questions by providing statistical insight on whether employees who receive a higher salary also receive more benefits.

Data we utilized for our study came from a dataset, Employee Compensation, from DataSF. The dataset provides details on different salaries adding up to a total salary and different benefits adding up to total benefits on each employee for the City of San Francisco. To address the issue of independence, we focused on the year of 2024, as many employees are working for the city during multiple years.

The Data section gives further details on the data that was utilized. Details on the statistical methods utilized are listed in the Methods section. Our statistical findings are listed in the Results section. A conclusion can be found in the Discussion section.

Data

The dataset we utilized, Employee Compensation (*Employee Compensation* 2013), provides data on the salary and benefits of each individual city employee since the fiscal year of 2013. The data was compiled and provided by the City Controller's Office. There are a total of 1.05 million data points in this dataset, where each row reflects a city employee. Variables that were relevant to our study included Total.Salary, which was each employee's annual salary and Total.Benefits, which was each employee's total benefits in monetary value. To allow numerical computations, all commas were removed from the data. To avoid dependence as many employees may be hired by the city in multiple years, we focused on the year of 2024, the most recent, which contains 86698 employees.

Table 1: Summary Statistics for Total Salary and Total Benefits (Year 2024)

	Value
Mean_Salary	117786.91
Salary_SD	80814.07
Min_Salary	-23666.55
Max_Salary	722186.26
Mean_Benefits	37339.22
Benefits_SD	22431.71

	Value
Min_Benefits	-17860.70
Max_Benefits	208640.04

Methods

A linear regression model was fitted based on the dataset using the lm() function in R (R Core Team 2024). A scatterplot depicting the predictor and response variables was also generated using the ggplot package (Wickham 2016). Our model is as follows:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

In our regression model above, the total salary that each employee, i, received is represented as the predictor variable (X), and the response variable (Y) was the total benefits that individual employee received. β_0 is the intercept, which is the expected value for the benefits recevied when the hypothetical salary is 0, β_1 is the slope, or expected change in benefits for every increase in dollar of salary, and ϵ_i is the error, which includes variations in our response variable (Y), the benefits, that are influenced by factors other than salary.

Analyzing the model and the estimated slope would show us how much a unit change in salary affects benefits. Running a t-test, our hypothesis test, would allow us to conclude whether or not the benefits received, our response variable, is related to the salary received, our predictor variable.

A limitation we have, however, is that we are not able to conclude causation between the two variables. For example, the department a that city employee works in could simply pay higher and give more benefits than other departments. In that sense, the benefits received is not caused by the total salary received, but is caused by the department the employee works in. Another example would include that some city workers are unionized, which would increase their benefits. In that sense, the benefits they receive is not caused by their salary, but by their union membership. Furthermore, our dataset only contains data on employees of the City of San Francisco. Trends may very likely differ if we look at data in another city or a private organization, a confounding variable.

We also looked into the assumptions of a standard linear regression model where the errors are assumed to have constant variance and are independent of one another. In this study, constant variance among the errors would indicate that the variation in benefits received, that is not explained by salary, is constant among all employees. The independence of errors would indicate that the amount of unexplained variation in the benefits of one employee is not affected by the amount of unexplained variation in benefits for another employee. While it was examined, in our situation the normality of errors would not have much of an effect as our sample size is significantly large. In addition to the assumptions on the errors, we also examined whether the relationship between total salary and total benefits was approximately linear, as required by a linear regression model.

Results

We ran a t-test for our hypothesis test where we have a null hypothesis, H_0, assuming that β_1, the slope, was equal to 0, indicating no relationship between the salary earned and the total benefits received. Our alternative hypothesis, H_a, would conclude that β_1, the slope, was not equal to 0, indicating that the total benefits received was related to the salary of an employee. To make a conclusion, we examined the p-value and compared it with our test size of 0.05. The test statistic that we utilized was our t-value:

$$t = \frac{b_1 - 0}{SE(b_1)}$$

We observed a p-value of less than $2 * 10^{-16}$. Based on the p-value being less than our test size of 0.05, we reject the null hypothesis for our hypothesis test and conclude that there is a relationship between the two variables. Our R^2 value of 0.7086 also indicates that total salary explains a lot of the variability in the total benefits received. The model also estimated slope, b_1, which shows that for every extra dollar of salary an employee gets, they get about \$0.27 in benefits.

Figure 1: Total Benefits vs. Total Salary for City of San Francisco Employees in 2024

To analyze whether or not our model follows the assumption that the regression function is linear and the errors have equal variance and are normally distributed, we examine the residual vs. predictor plot and the QQ plot. Given that we have individual and unrelated employees from the year of 2024, we can conclude that the errors are independent from each other.

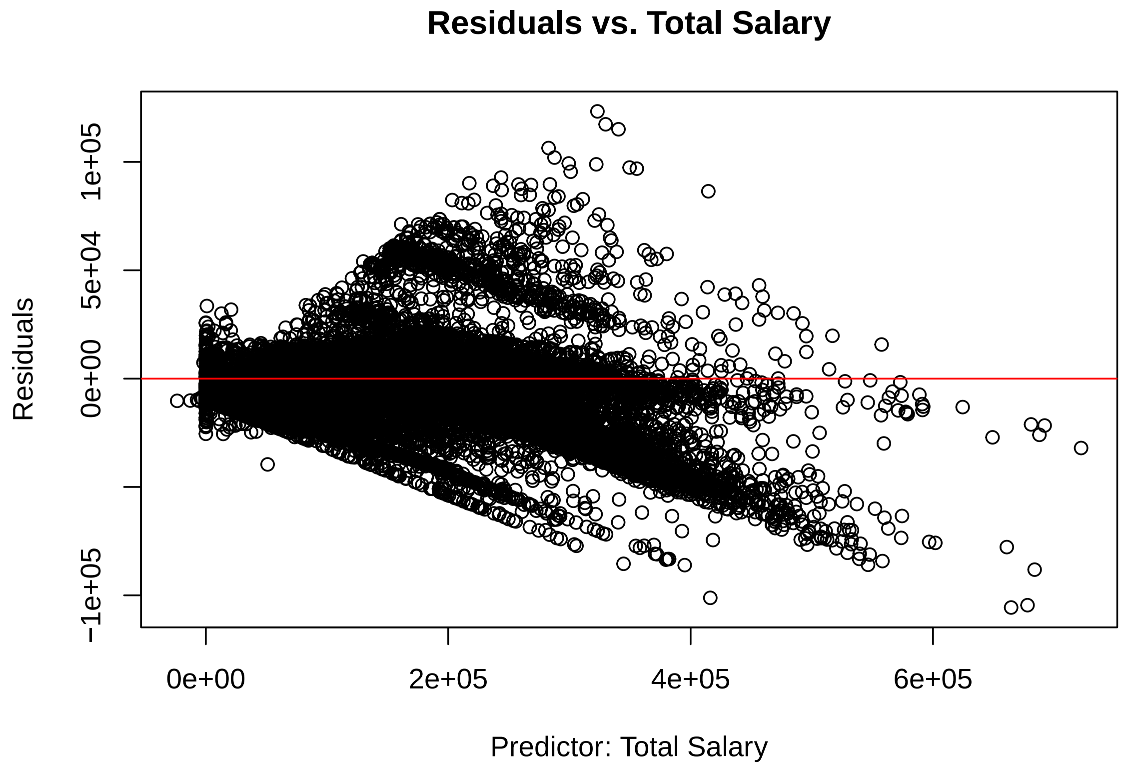

Figure 2: Residuals vs. Total Salary (Predictor), Heteroscedasticity inidicates non-constant variances among residuals

Upon examining the residual vs predictor plot, we can see a funnel shape as the salary increases. This shows that there is a higher variance among higher salaries and that the errors are not of constant variance. However, there is no “u-shape” or “wave” that would indicate a non-linear relationship.

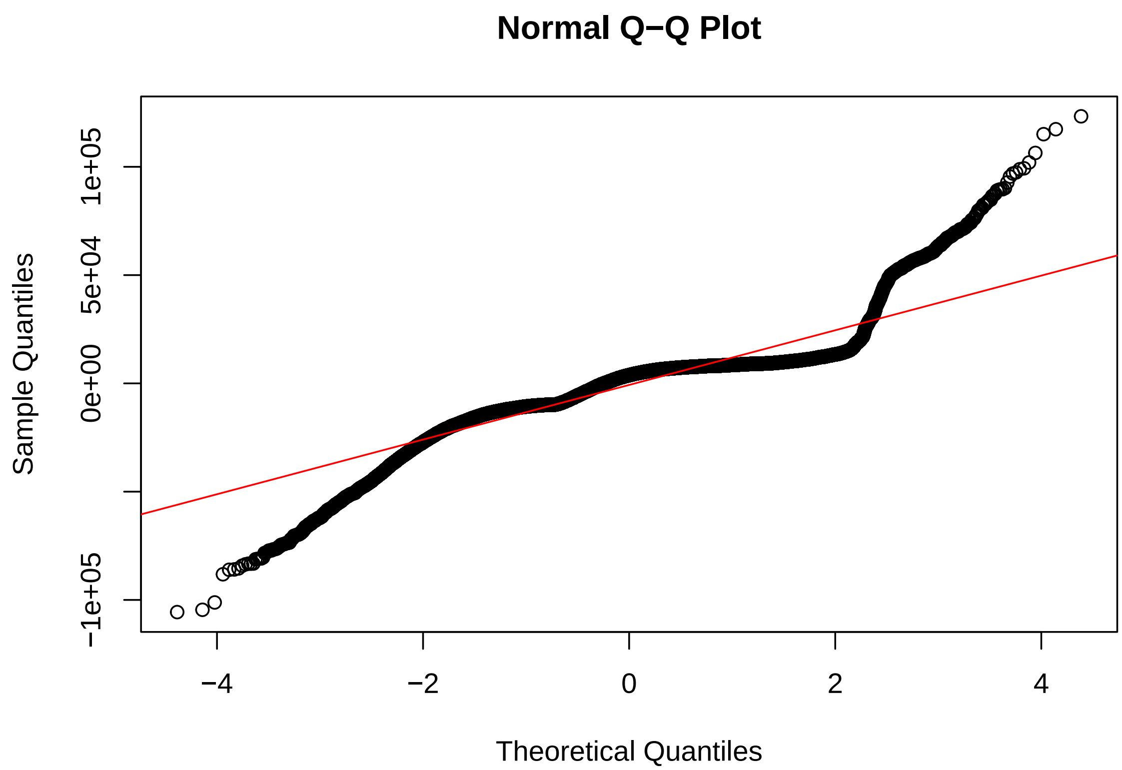

Figure 3: QQ-Plot, Compares Errors To A Normal Distribution, Heavy Tails Reflect Not Normally Distributed

Upon examining the Q-Q, or Quantile-Quantile, plot, we can clearly see that the distribution of residuals has heavier tails on both the right and left tails. Thus, we conclude our residuals are not normally distributed. However, due to our large sample size of 86698, normality would not have much of an effect.

Discussion

To determine whether or not the benefits an employee receives is related to their salary, we analyzed the total annual salary and total benefit value of 86698 employees working for the City of San Francisco in 2024. We considered the total salary to be the predictor variable and the total benefits value to be our response variable when fitting our regression model.

To determine whether or not the salary received is related to benefits obtained, we ran a t-test as our hypothesis test with a test size of 0.05 where we considered a null hypothesis that indicates no relation between our response and predictor variables and an alternative hypothesis that indicates our response and predictor variables are related. In our study, that null hypothesis would be concluding that the amount of benefits an employee receives has no relation to the salary they get. The alternative hypothesis would conclude that the amount of benefits an employee receives is related to their salary. To determine how much a change in salary would impact benefits we looked into the estimated slope for the salary, our predictor variable, in our model.

Considering a two-sided t-test with a test size of 0.05, we observed a p-value that is significantly lower than 0.05, thus we reject the null hypothesis and concluded that our variables are related with a positive correlation. A high R^2, shows that variance within the benefits received is explained by the salary received, especially in a large sample size of 86698 city employees. Thus based on our study, we conclude that the amount of benefits an employee receives is related to the salary an employee receives. Analyzing our model, we also found an estimated slope of 0.2337 for our total annual salary. This shows that for every unit increase in salary, an employee gets around \$0.23 more in benefits.

However, this research has many limitations. One of which is in the sense of causal relationships. Although we can clearly see that the total benefits an employee receives is positively correlated with the total annual salary they receive, we cannot say a high salary "causes" a high amount of benefits. As mentioned above, a particular department an employee works for could have significantly more funding. Should an employee work for that particular department, they would have higher annual salary and benefits received. Many city employees are members of unions. Unions would generally negotiate better pay and benefits for their members. Lastly, our data focuses on employees who work for the city of San Francisco. Trends that we see here may not be applicable for employees who work for other cities or non-governmental workers.

Another limitation comes with that fact that different departments may calculate benefits very differently. Our model assumes that the relationship between salary and benefits is consistent across all employees. This brings up the issue that a single global linear model may not accurately capture patterns that differ across departments. However, different departments were included to address the issue of our data being representative as we do not wish to overfit on one department.

Furthermore, by looking into the residual vs. predictor and Q-Q plots, we can conclude that our assumptions on equal variance, or variation in benefits not explained by the salary, and normality of the errors, have been violated. Due to a large sample size, the violation on normality would not have a significant effect. However, non-equal variance among errors could make our confidence intervals and standard errors off, requiring us to take our hypothesis test with a grain of salt where we could have falsely rejected our null hypothesis.

To make our findings more conclusive, instead of using a simple linear regression model where we only consider one predictor variable, the total salary received, we could utilize a multiple linear regression model where we have multiple predictor variables. Variables we mentioned above (e.g. union membership and department) could be one-hot encoded and included as additional predictor variables. The data on the estimated slopes of these predictor variables could be used to determine which predictor variable is more highly correlated with our response variable, the total benefits received. In addition, we could look into data on compensation from other cities or private organizations to get a sample that is more representative and is not limited to employees for the city of San Francisco. To address non-constant variances among the errors we could introduce variable transformations which would make our hypothesis test more reliable.

References

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